

ECE 341
Homework 3

The problem numbers listed below are from the Lathi textbook, 3rd edition.

Prob. Number	Course Outcome	Grade
3.1-1 (a), (c)	1-b, 7-c	10
3.1-2 (b)	1-b, 7-c	10
3.2-7 (c), (d), (e)	1-b, 7-c	15
3.3-3 (a), (b)	1-b, 7-c	10
3.4-1	1-b, 7-c	10
3.4-12	1-b, 7-c	15
TOTAL:		70

1-b. Evaluate and classify signals to identify unstable systems.

7-c. Apply new knowledge to understand continuous and discrete systems.

3.1.1 a) Taking the individual values of each non-zero $x[n]$ shown in the the figure,

$$\sum_{n=-2}^0 \left(\frac{3}{2}n + 3\right)^2 + \sum_{n=1}^2 \left(-\frac{3}{2}n + 3\right)^2 = 1^2 + 2^2 + 3^2 + 2^2 + 1^2 = 19.$$

b) Again, taking the individual values of each non-zero $x[n]$,

$$\sum_{n=-3}^3 \left(\frac{9}{3}n\right)^2 = (9 + 9) + (36 + 36) + (81 + 81) = 252.$$

3.1.2 a) Finding the power of the signal, averaged across its period,

$$\frac{1}{12}(2 \times 3^2) + (2 \times 2^2) + (2 \times 1^2) \approx 2.33.$$

3.2.7 a) $x[n - 6]$ is the signal $x[n]$ time-shifted forward by 6 units, and may be represented as

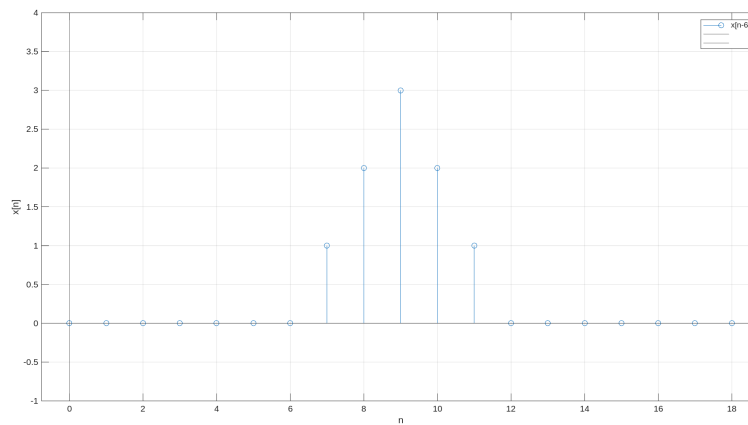


Figure 1: $x[n - 6]$

b) $x[3n]$ is the compression of the signal $x[n]$ by a factor of 3 units, and may be represented as

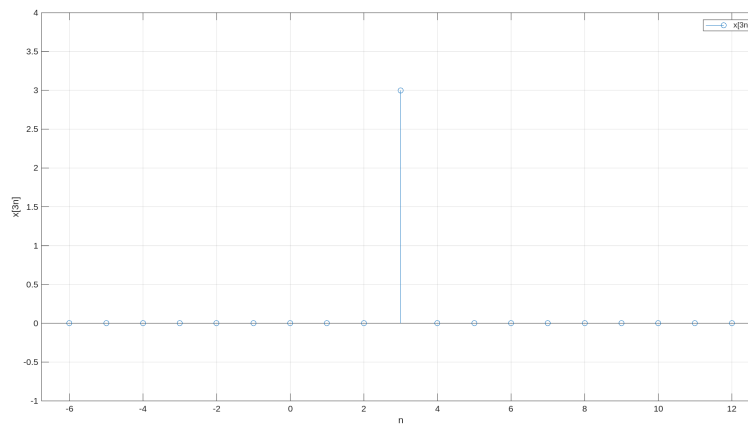
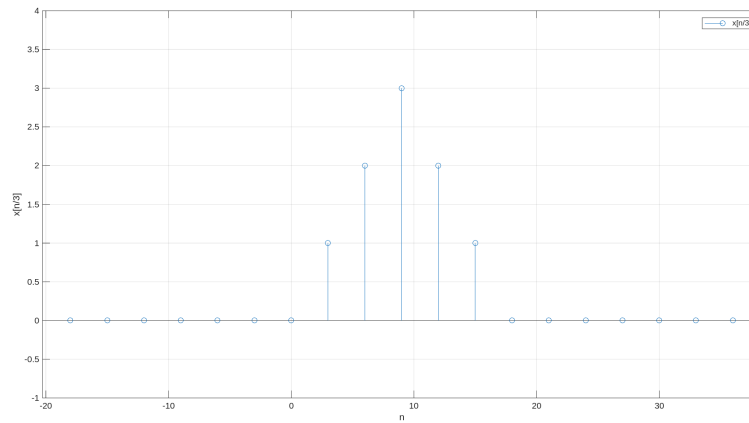
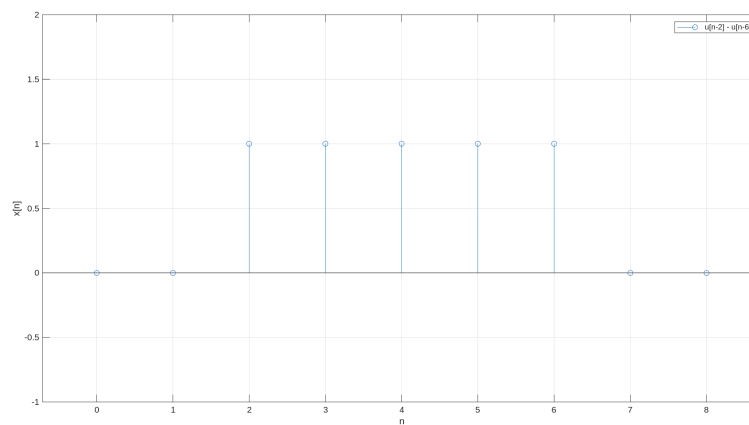


Figure 2: $x[3n]$

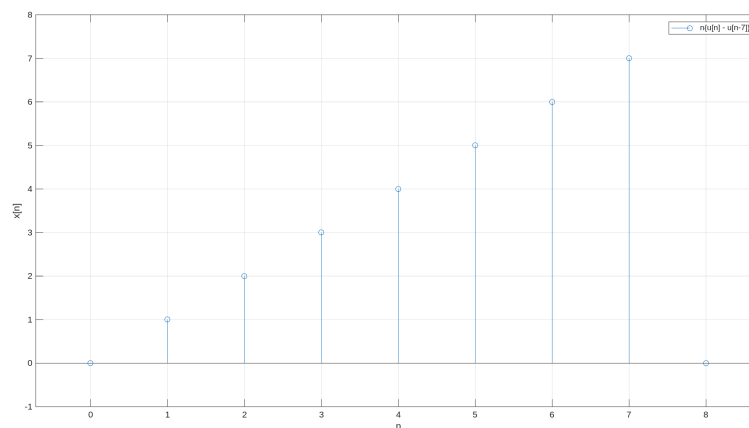
c) $x[\frac{n}{3}]$ is the expansion of the signal $x[n]$ by a factor of 3 units, and may be represented as

Figure 3: $x[\frac{n}{3}]$

3.3.3 a) The signal $u[n-2] - u[n-6]$ may be represented as

Figure 4: $u[n-2] - u[n-6]$

b) The signal $n\{u[n] - u[n-7]\}$ may be represented as

Figure 5: $n\{u[n] - u[n-7]\}$

3.4.1 a) Since the value of $y[n]$ corresponds to the value of the current item, plus the cumulative value of

all previous items, so that

$$y[n] - y[n-1] = x[n].$$

- b) The system can be represented graphically as a block diagram

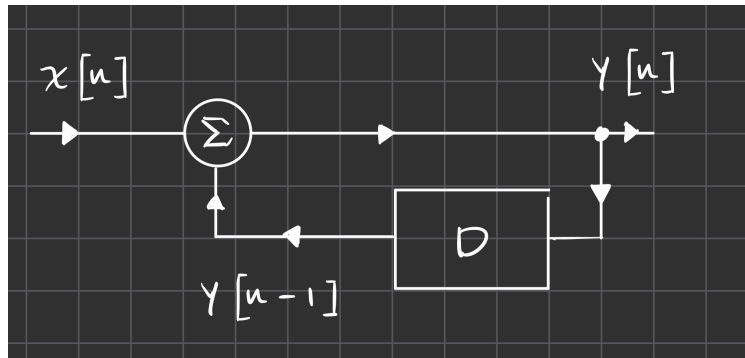


Figure 6: Block diagram system representation

- 3.4.12 a) BIBO stability is determined based on whether the input is bounded for a bounded output. In this example, if the bounded input $x[n-1] = 0$, $y[n+1]$ approaches infinity, and so the system is BIBO-unstable.
- b) Adjusting the system to be in *delay* form,

$$y[n] = \frac{x[n-1]}{x[n]},$$

we can see clearly that the system relies on inputs prior to the present input at $n = 0$, and so the system is **not** memoryless.

- c) Since the system does not depend on **future** values of $x[n+1]$, the system can be considered **causal**.